

Statistical Inference

Labix

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Abstract

Notes for the basics of Probability Theory.

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1 Fundamentals of Statistics

1.1 Statistical Models

Definition 1.1.1 (Statistical Models)

Let $(\Omega, \mathcal{F})$ be a measurable space. A statistical model is a family of probability measures

$$\mathcal{P} = \{P_\theta : \mathcal{F} \rightarrow [0, 1] \mid \theta \in I\}$$

for some indexing set I .

- I is called the parameter space of the statistical model.
- We denote the corresponding probability distribution by f_θ .

Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. For each probability measure P_θ , the probability density function f_X also depends on θ , and we obtain a family of probability density functions $\{f_\theta \mid \theta \in I\}$. Some authors use this as the definition of statistical models, but our definition naturally implies this alternative one.

Definition 1.1.2 (Parametric Models)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P}$ be a statistical model indexed by I . We say that $\mathcal{P}$ is parametric if $I \subseteq \mathbb{R}^k$ for some $k \in \mathbb{N}$.

1.2 Random Samples and Statistics

Definition 1.2.1 (Random Samples)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be random variables. We say that they are a random sample of size n from the population f if the following are true.

- $X_1, \dots, X_n$ are pairwise independent.
- The probability density (mass) function is given by $f_{X_i} = f$.

The fact that their probability density functions are equal imply that the random variables are identically distributed.

Definition 1.2.2 (Statistics)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. A statistic of the random sample is a random variable $T(X_1, \dots, X_n)$ for some function $T : \mathbb{R}^n \rightarrow \mathbb{R}$.

Definition 1.2.3 (Sample Mean)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Define the sample mean of the random sample to be the statistic defined by

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

By the weak law of large numbers, we know that $\bar{X}$ converges in mean square to the mean of the random sample.

Definition 1.2.4 (Sample Variance)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Define the sample variance of the random sample to be the statistic defined by

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Definition 1.2.5 (Sample Standard Deviation)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Define the sample standard deviation of the random sample to be the statistic defined by

$$S = \sqrt{S^2}$$

2 Principles of Data Reduction

2.1 Sufficiency Principle

A sufficient statistic T is a statistic where, informally speaking, all information about P must come from T .

Definition 2.1.1 (Sufficient Statistics)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Let $T : \mathbb{R}^n \rightarrow \mathbb{R}$ be a statistic. We say that T is sufficient for I if the conditional probability density function $f_{X_1, \dots, X_n | T}$ is a constant function with respect to θ .

Example 2.1.2

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample of Bernoulli distribution with parameter $\theta \in (0, 1)$ (Note that for each θ , $P_{X_i} = f_{X_i}$ is different since the formula for f_{X_i} depends on θ). Then $T = \sum_{i=1}^n X_i$ is a sufficient statistic for θ .

Proof

Let $\theta \in \mathbb{R}$. We use the above lemma. We have

$$\begin{aligned} P_\theta(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = T(\mathbf{x})) &= \frac{f_{X_1, \dots, X_n}(x_1, \dots, x_n)}{f_{T(X_1, \dots, X_n)}(x_1, \dots, x_n)} \\ &= \frac{\prod_{i=1}^n f_{X_i}(x_i)}{f_{X_1 + \dots + X_n}(x_1, \dots, x_n)} \\ &= \frac{\prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i}}{\binom{n}{\sum_{i=1}^n x_i} \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}} \\ &= \frac{\theta^{\sum_{i=1}^n x_i} (1 - \theta)^{\sum_{i=1}^n x_i}}{\binom{n}{\sum_{i=1}^n x_i} \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}} \\ &= \frac{1}{\binom{n}{\sum_{i=1}^n x_i}} \end{aligned}$$

■

Theorem 2.1.3 (Neyman-Fisher Factorization Theorem)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Let $T : \mathbb{R}^n \rightarrow \mathbb{R}$ be a statistic. Then T is sufficient for I if and only if there exists $g : \mathbb{R} \times I \rightarrow \mathbb{R}$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$f_\theta(x) = g(T(x), \theta)h(x)$$

for all $x \in \text{im}(X)$ and all $\theta \in I$.

Definition 2.1.4 (Minimal Sufficient Statistics)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $\{D_\theta \mid \theta \in I\}$ be a family of distributions. Let $X_\theta : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution D_θ . Let T be a sufficient statistic of the random sample. We say that T is minimal if for any other sufficient statistic T' , $T'(x) = T'(y)$ implies $T(x) = T(y)$.

Definition 2.1.5 (Ancillary Statistics)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $\{D_\theta \mid \theta \in I\}$ be a family of distributions. Let $X_\theta : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution D_θ . Let T be a statistic of the random sample. We say that T is ancillary if T is a constant function with respect to θ .

Definition 2.1.6 (Complete Statistics)

Theorem 2.1.7 (Basu's Theorem)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $\{D_\theta \mid \theta \in I\}$ be a family of distributions. Let $X_\theta : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution D_θ . Let T be a statistic of the random sample. If T is complete and minimal sufficient, then T is independent of any ancillary statistic.

2.2 Likelihood Principle

Definition 2.2.1 (The Likelihood Function)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Define the likelihood function $L : I \rightarrow \{g : \mathbb{R} \rightarrow \mathbb{R} \mid g \text{ is a function}\}$ of the random sample to be

$$L(\theta) = f_\theta$$

2.3 Conditionality Principle

2.4 Equivariance Principle

3 Point Estimators

Suppose some real life problem is modelling statistically by the probability space $(\Omega, \mathcal{F}, P)$. To probe what P looks like, we sample the space using random variables $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$. Based on the sample, we may infer that P may have parameter $\hat{\theta}$ (naturally $\hat{\theta}$ should be a function of $X_1, \dots, X_n$). We may then need to evaluate how accurate $\hat{\theta}$ is compared to the actual parameter θ .

There are two main branches in estimator theory: We need different methods for finding estimators, and we need to evaluate how accurate the estimators are.

There are also two types of estimators: Point estimators and interval estimators.

3.1 Basics of Point Estimators

Definition 3.1.1 (Point Estimators)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Let $T : \mathbb{R}^n \rightarrow \mathbb{R}$ be a statistic. We say that T is a point estimator if $\text{im}(T) \subseteq I$.

Estimators: A way to predict what the true parameter is (they are random variables because it depends on the actual measurements). Estimate: The value of an estimator of a given measurement. We also need a way to evaluate how good the estimators are (their input is a random variable and output is a real number). (In particular, one way to get a good estimator is to first find a method of evaluating estimators, then minimize it)

3.2 Evaluating the Accuracy of Point Estimators

Definition 3.2.1 (Bias of a Point Estimator)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Let $\hat{\theta} : \mathbb{R}^n \rightarrow I$ be a point estimator. Define the bias of $\hat{\theta}$ to be

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

Definition 3.2.2 (Unbiased Point Estimators)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Let $\hat{\theta} : \mathbb{R}^n \rightarrow I$ be a point estimator. We say that $\hat{\theta}$ is unbiased if $\text{Bias}(\hat{\theta}) = 0$.

Definition 3.2.3 (Mean Squared Error)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Let $\hat{\theta} : \mathbb{R}^n \rightarrow I$ be a point estimator. Define the mean squared error of $\hat{\theta}$ to be

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

Theorem 3.2.4 (The Bias Variance Trade Off)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \rightarrow \mathbb{R}$ be a random sample. Let $\hat{\theta}$ be a point estimator of the random sample. Then we have

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2$$

3.3 Constructing Point Estimators

Definition 3.3.1 (Method of Moments)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $J_1, \dots, J_k \subseteq \mathbb{R}$ be indexing sets. Let $\{D_{\theta_1, \dots, \theta_k} \mid \theta_i \in J_i\}$ be a family of distributions. Let $X_{\theta_1, \dots, \theta_k} : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution $D_{\theta_1, \dots, \theta_k}$ and probability distribution function $f_{\theta_1, \dots, \theta_k}$. Define the methods of moments estimator to be the statistics $\hat{\theta}_1, \dots, \hat{\theta}_k$ given by the solving the system of equations

$$\frac{1}{n} \sum_{j=1}^n (X_{\theta_1, \dots, \theta_k})_j^i = E[D_{\theta_1, \dots, \theta_k}^i]$$

for $1 \leq i \leq k$.

Example 3.3.2

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with normal distribution $X_i \sim \text{Normal}(\theta_1, \theta_2)$. The method of moments estimator for X is given by solving

$$\frac{1}{n} \sum_{j=1}^n (X_{\theta_1, \dots, \theta_k})_j = E[\text{Normal}(\hat{\theta}_1, \hat{\theta}_2)] \quad \text{and} \quad \frac{1}{n} \sum_{j=1}^n (X_{\theta_1, \dots, \theta_k})_j^2 = E[\text{Normal}(\hat{\theta}_1, \hat{\theta}_2)^2]$$

which we may rewrite as

$$\overline{X_{\theta_1, \dots, \theta_k}} = \hat{\theta}_1 \quad \text{and} \quad \frac{1}{n} \sum_{j=1}^n (X_{\theta_1, \dots, \theta_k})_j^2 = \hat{\theta}_1^2 + \hat{\theta}_2^2$$

Omitting the dependency of the random variables on the parameters, we get

$$\hat{\theta}_1 = \bar{X} \quad \text{and} \quad \hat{\theta}_2 = \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})^2$$

We are lucky in the above example because the estimator accurately recovers the actual parameters. But in general, this is not true.

Example 3.3.3

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with binomial distribution $X_i \sim \text{Binom}(\theta_1, \theta_2)$. The method of moments estimator for X is given by solving

$$\frac{1}{n} \sum_{j=1}^n X_j = E[\text{Binom}(\hat{\theta}_1, \hat{\theta}_2)] \quad \text{and} \quad \frac{1}{n} \sum_{j=1}^n X_j^2 = E[\text{Binom}(\hat{\theta}_1, \hat{\theta}_2)^2]$$

which we may rewrite as

$$\bar{X} = \hat{\theta}_1 \hat{\theta}_2 \quad \text{and} \quad \frac{1}{n} \sum_{j=1}^n X_j^2 = \hat{\theta}_1 \hat{\theta}_2 (1 - \hat{\theta}_2) + \hat{\theta}_1^2 \hat{\theta}_2^2$$

Solving the simultaneous equation gives

$$\hat{\theta}_1 = \frac{\bar{X}^2}{\bar{X} - \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})^2} \quad \text{and} \quad \hat{\theta}_2 = \frac{\bar{X} - \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})^2}{\bar{X}} = \frac{\bar{X}}{\hat{\theta}_1}$$

Definition 3.3.4 (Maximum Likelihood Estimators)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Let $\hat{\theta} : \mathbb{R}^n \rightarrow I$ be a point estimator. We say that $\hat{\theta}$ is a

maximum likelihood estimator if there exists $x \in \mathbb{R}^n$ such that

$$L(\hat{\theta}(x))(x) = \max\{L(\theta)(x) : \theta \in I\}$$

Lemma 3.3.5

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . If L is differentiable, then the maximum likelihood estimator exists and is given by the root of the equation

$$\nabla L = 0$$

4 Hypothesis Testing

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. We are interested in the following question: Suppose we know that the parameter lies in one of two set $H_0, H_1 \subseteq I$ with $H_0 \cap H_1 = \emptyset$. We wish to use random samples $X : \Omega \rightarrow \mathbb{R}^n$ to determine whether it lies in H_0 or H_1 .

4.1 Basics of Hypothesis Testing

Definition 4.1.1 (Hypothesis)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. A hypothesis of the parameter are two subsets $H_0, H_1 \subseteq I$ such that $H_0 \cap H_1 = \emptyset$.

Definition 4.1.2 (Test Functions)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . A test function is a function $\delta : \text{im}(X) \rightarrow \{0, 1\}$.

A test function takes the value 0 when the sample indicates that the parameter lies in H_0 , and vice versa for H_1 . A test function usually take values depending on a certain condition, so they are usually of the form

$$\delta(x) = \begin{cases} 1 & \text{if } T(x) \in C \\ 0 & \text{otherwise} \end{cases}$$

for some statistic T and some set $C \in \text{im}(T)$.

4.2 Evaluating the Accuracy of Hypothesis Tests

Similar to point estimators, we also need to evaluate how accurate the test function δ is. In the case of hypothesis testing, we can easily classify the possible errors.

- True parameter: 0, decision: 0. There is no error.
- True parameter: 0, decision: 1. We call this type I error.
- True parameter: 1, decision: 0. We call this type II error.
- True parameter 1, decision: 1. There is no error.

Definition 4.2.1 (Error Probabilities)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Let $H_0, H_1 \subseteq I$ be a hypothesis. Let $\delta : \text{im}(X) \rightarrow \{0, 1\}$ be a test function.

- Define the type I error probability to be the map $h_0 : H_0 \rightarrow [0, 1]$ given by

$$h_0(\theta) = P_\theta(\delta = 1)$$

- Define the type II error probability to be the map $h_1 : H_1 \rightarrow [0, 1]$ given by

$$h_1(\theta) = P_\theta(\delta = 0)$$

4.3 Constructing Test Functions

It is not possible to simultaneously minimize h_0 and h_1 . Therefore one way to find a good test function is to put emphasis on H_0 before H_1 . In practice most scientific experiments have a more important error to prevent from having false positives.

Definition 4.3.1 (The Neyman-Pearson Framework)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model. Let $X : \Omega \rightarrow \mathbb{R}^n$

be a random sample with distribution f_θ . Let $H_0, H_1 \subseteq I$ be a hypothesis. The Neyman-Pearson framework is the following algorithm that ???

1. Fix $\alpha \in (0, 1)$ and call it the significance level of the test.
2. Define the class of test functions that respect the level

$$\mathcal{D}(H_0, \alpha) = \{\delta : \text{im}(X) \rightarrow \{0, 1\} \mid \sup_{\theta \in H_0} P_\theta(\delta = 1) \leq \alpha\}$$

3. Find the test function $\delta \in \mathcal{D}(H_0, \alpha)$ that minimizes h_1 .

4.4 Hypothesis Testing for One Dimensional Parameters

Definition 4.4.1 (Simple vs Lateral Hypothesis)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model such that $I \subseteq \mathbb{R}$. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Let $H_0, H_1 \subseteq I$ be a hypothesis.

- We say that the hypothesis is a simple vs simple if H_0, H_1 are one point sets.
- We say that the hypothesis is unilateral if $H_0 = \{\theta \in I \mid \theta \leq \theta_0\}$ and $H_1 = I \setminus H_0$ or $H_0 = \{\theta \in I \mid \theta \geq \theta_0\}$ and $H_1 = I \setminus H_0$ for some $\theta_0 \in I$.
- We say that the hypothesis is simple vs bilateral if H_0 is a one point set and $H_1 = I \setminus H_0$.

Definition 4.4.2 (p-Value)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model such that $I \subseteq \mathbb{R}$. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Let $H_0, H_1 \subseteq I$ be a hypothesis that it is either simple vs simple, unilateral or simple vs bilateral. Let $\delta : \text{im}(X) \rightarrow \{0, 1\}$ be a test function of one of the following forms:

$$\delta = 1_{T > q_{1-\alpha}} \quad \text{or} \quad \delta = 1_{T \leq q_\alpha}$$

where T is some statistic and q_z is the z -quantile of the distribution $G_0(t) = P_{\theta_0}(T \leq t)$. Define the p -value of X to be

$$\inf\{\alpha \in (0, 1) \mid \delta_\alpha = 1\}$$

5 Interval Estimators

5.1 Basics of Interval Estimators

Definition 5.1.1 (Two Sided Confidence Intervals)

Let $(\Omega, \mathcal{F})$ be a measurable space. Let $\mathcal{P} = \{P_\theta \mid \theta \in I\}$ be a parametric model such that $I \subseteq \mathbb{R}$. Let $X : \Omega \rightarrow \mathbb{R}^n$ be a random sample with distribution f_θ . Let $\alpha \in (0, 1)$. Let $L, U : \mathbb{R}^n \rightarrow \mathbb{R}$ be two statistics such that

$$\inf_{\theta \in I} P_\theta(L \leq \theta \leq U) \geq 1 - \alpha$$

Define the two sided confidence interval for θ with confidence level $1 - \alpha$ to be the interval

$$[L(X), U(X)] \subseteq \mathbb{R}$$

6

6.1

Let X, Y be random variables. Regression is about understanding the relationship of Y given X . Clearly this is a question in conditional probability: If I observed $X = x$, what can I say about the distribution of Y ?

Suppose we sample p data points $(X_1, Y_1), \dots, (X_p, Y_p)$, i.e. we assume we have a random sample with distribution (X, Y) . We would like to assume that there is some function f such that when $X_i = x_i$ and $Y_i = y_i$ are observed, then we have $y_i = f(x_i)$. In reality, if we also consider the fact that our observations may have measurement errors or inherent randomness, then we find that $y_i \in f(x_i) + \varepsilon$ where ε is some random variable representing the measurement error / inherent randomness. Hence $Y_i = f(x_i) + \varepsilon$ is really the starting equation for us to use to find what f is. Assume that $E[\varepsilon] = 0$, that is on average our measurement error is 0, we find that $E[Y_i] = f(x_i)$. Since this is a function of x_i , we may rewrite it as $E[Y_i | X_i = x_i] = f(x_i)$. This means that we are asking the question of on average, what is the relationship between the outcome Y_i given $X_i = x_i$.

The above assumes that the relationship between Y and X is given by a function. In general, there are other types of relationships that we may ask for e.g. variance regression: asking for how the variance of Y is related to X instead of how the outcome Y is related to X .

Definition 6.1.1 (Regression Models)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $(X_1, Y_1), \dots, (X_p, Y_p) : \Omega \rightarrow \mathbb{R}^n \times \mathbb{R}$ be a random sample. A regression model is a parametric statistical model $\mathcal{P}$ on the random variable $Y_i | X_i = x$.

Notice that i does not matter in the regression model because (X_i, Y_i) are independently identically distributed.

Definition 6.1.2 (Linear Regression Models)

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $(X_1, Y_1), \dots, (X_p, Y_p) : \Omega \rightarrow \mathbb{R}^n \times \mathbb{R}$ be a random sample. Let $\mathcal{P}$ be a regression model parameterized by $I \subseteq \mathbb{R}^k$. We say that $\mathcal{P}$ is linear if $E[Y_i | X_i = x_i]$ is a linear function of the parameter $\theta \in I$.

Definition 6.1.3 (Design Matrix)